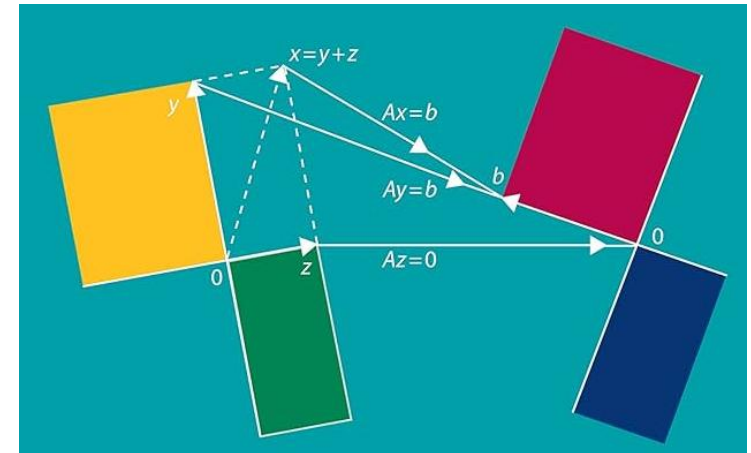


Eigenvalues and Eigenvectors

(Linear Algebra)

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When a square matrix A acts upon a vector x , it generally outputs a new vector Ax . Usually this new vector will be stretched and rotated. For example, if we take the matrix

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

and apply it to the vector $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ we get:

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

If we apply A to the vector $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ we get:

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

if we apply A to the vector $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ we get:

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

The matrix A leaves the vector $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ unchanged! In particular, it does not rotate the vector. When a matrix A acts upon a vector and does not rotate it, we have $A\mathbf{x} = \lambda\mathbf{x}$, where λ is a scaling factor. In our example $\lambda = 1$. We call such a vector an *eigenvector* for the matrix A , and the associated scaling factor λ an *eigenvalue*.

Definition

Let A be an $n \times n$ matrix. If there exist a real value λ and a non-zero $n \times 1$ vector x satisfying,

$$Ax = \lambda x$$

then

we refer to λ as an eigenvalue of A , and x as an eigenvector of A corresponding to λ .

The basic equation for eigenvectors and eigenvalues is

$$A\mathbf{x} = \lambda\mathbf{x}$$

where A is a matrix and λ is a number. One property we see right away is

$$A^2\mathbf{x} = \lambda A\mathbf{x} = \lambda^2\mathbf{x}$$

and in general

$$A^n\mathbf{x} = \lambda^n\mathbf{x}.$$

So, the eigenvectors of A^n are the same as the eigenvectors of A , while the eigenvalues for A^n are the eigenvalues for A raised to the n th power.

Reveal Important Directions in Data

- Suppose students are described by Height and Weight
- These two features are often correlated.
- Instead of studying both separately, eigenvectors identify the principal direction where the data varies the most.
- This is the foundation of Principal Component Analysis (PCA).

Dimensionality Reduction

- Large datasets may contain hundreds or thousands of variables.
- Eigenvectors help reduce them while preserving most of the information.
- Original data: 1000 features
- After PCA: 20 principal components
- Much faster machine learning with little loss of information.

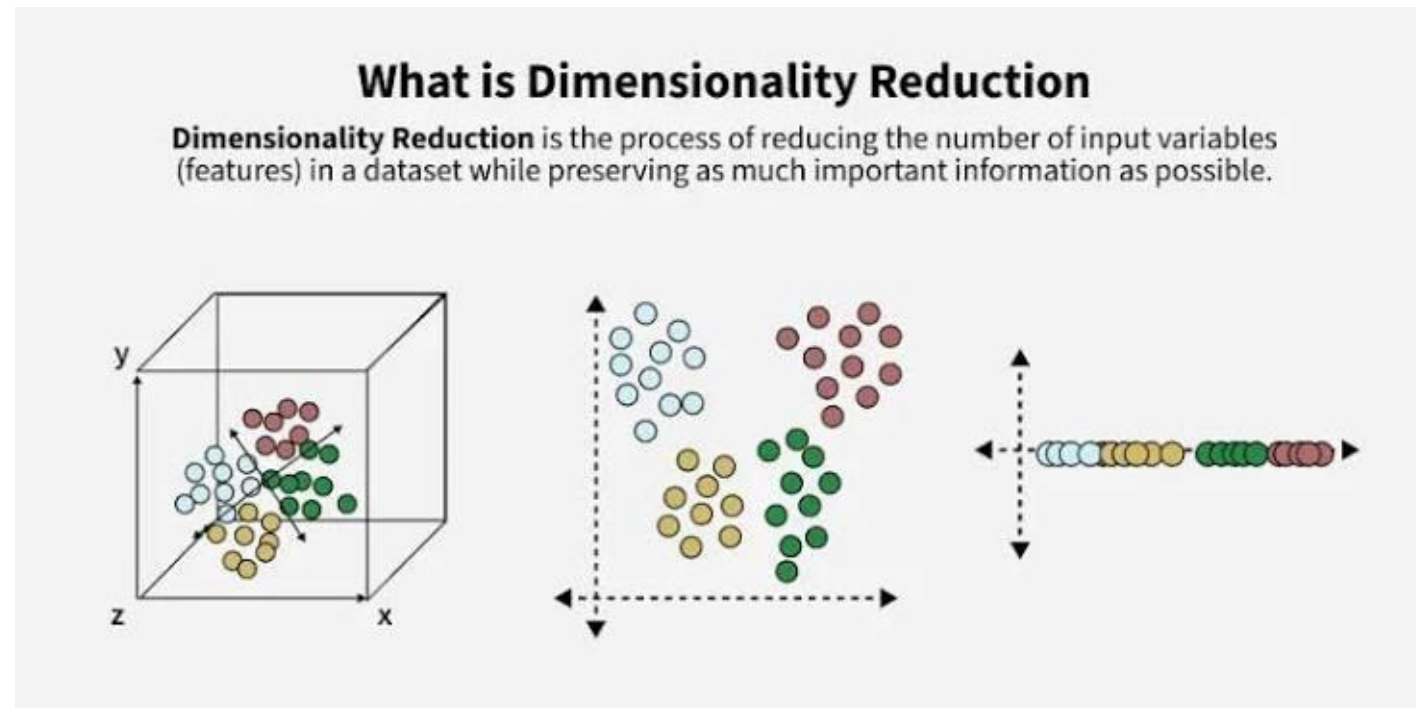


Image Compression

- Digital images contain many pixels.
- Eigenvectors identify the most important patterns.
- Keeping only the largest eigenvalues allows us to reconstruct images using much less storage.



Three levels of JPG compression. The left-most image is the original. The middle image offers a medium compression, which may not be immediately obvious to the naked eye without closer inspection. The right-most image is maximally compressed.

Purpose of Learning in Computer Science

- Data Compression and Dimensionality Reduction:
 - reduce the dimensionality of large datasets, increasing interpretability while minimizing information loss.
- Machine Learning:
 - Feature Extraction and Selection
- Computer Vision and Image Processing:
 - Image Compression and Face Recognition
- Natural Language Processing (NLP):
 - Word Embedding

How to calculate?

$$A\mathbf{x} = \lambda\mathbf{x}$$

then

$$(A - \lambda I)\mathbf{x} = \mathbf{0}$$

So, the matrix $(A - \lambda I)$ has a nontrivial nullspace, and therefore must be singular. So,

$$\det(A - \lambda I) = 0.$$

So, if λ is an eigenvalue of A then $\det(A - \lambda I) = 0$. It turns out that this equation can be used to calculate *every* eigenvalue. The equation $\det(A - \lambda I) = 0$ will be a polynomial equation in λ , and its roots will give us all the eigenvalues. We call this polynomial equation the *characteristic equation* of A .

Eigenvalues first. If $(A - \lambda I)x = 0$ has a nonzero solution, $A - \lambda I$ is not invertible. ***The determinant of $A - \lambda I$ must be zero.*** This is how to recognize an eigenvalue λ :

Eigenvalues The number λ is an eigenvalue of A if and only if $A - \lambda I$ is singular.

Equation for the n eigenvalues of A

$$\det(A - \lambda I) = 0. \quad (5)$$

This “*characteristic polynomial*” $\det(A - \lambda I)$ involves only λ , not x . Since λ appears all along the main diagonal of $A - \lambda I$, the determinant in (5) includes $(-\lambda)^n$. Then equation (5) has n solutions λ_1 to λ_n and A has n eigenvalues.

An n by n matrix has n eigenvalues (repeated λ 's are possible !) Each λ leads to x :

For each eigenvalue λ solve $(A - \lambda I)x = 0$ or $Ax = \lambda x$ to find an eigenvector x .

Example 4 $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ is already singular (zero determinant). Find its λ 's and $\mathbf{x}$'s.

When A is singular, $\lambda = 0$ is one of the eigenvalues. The equation $A\mathbf{x} = 0\mathbf{x}$ has solutions. They are the eigenvectors for $\lambda = 0$. But $\det(A - \lambda I) = 0$ is the way to find *all* λ 's and $\mathbf{x}$'s. Always subtract λI from A :

Subtract λ along the diagonal to find $A - \lambda I = \begin{bmatrix} 1 - \lambda & 2 \\ 2 & 4 - \lambda \end{bmatrix}.$ (6)

Take the determinant “ $ad - bc$ ” of this 2 by 2 matrix. From $1 - \lambda$ times $4 - \lambda$, the “ ad ” part is $\lambda^2 - 5\lambda + 4$. The “ bc ” part, not containing λ , is 2 times 2.

$$\det \begin{bmatrix} 1 - \lambda & 2 \\ 2 & 4 - \lambda \end{bmatrix} = (1 - \lambda)(4 - \lambda) - (2)(2) = \lambda^2 - 5\lambda. \quad (7)$$

Set this determinant $\lambda^2 - 5\lambda$ to zero. One solution is $\lambda = 0$ (as expected, since A is singular). Factoring into λ times $\lambda - 5$, the other root is $\lambda = 5$:

$$\det(A - \lambda I) = \lambda^2 - 5\lambda = 0 \quad \text{yields the eigenvalues} \quad \lambda_1 = 0 \quad \text{and} \quad \lambda_2 = 5.$$

Now find the eigenvectors. Solve $(A - \lambda I)\mathbf{x} = \mathbf{0}$ separately for $\lambda_1 = 0$ and $\lambda_2 = 5$:

$$(A - 0I)\mathbf{x} = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \text{yields an eigenvector} \quad \begin{bmatrix} y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix} \quad \text{for } \lambda_1 = 0$$
$$(A - 5I)\mathbf{x} = \begin{bmatrix} -4 & 2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \text{yields an eigenvector} \quad \begin{bmatrix} y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \text{for } \lambda_2 = 5$$

Note that these eigenvectors are *not* unique. In fact, any non-zero multiple $c\mathbf{x}$ ($c \neq 0$) of an eigenvector is another eigenvector.

Oh, and we should probably mention right now that if $\mathbf{x} = \mathbf{0}$ then $A\mathbf{x} = \lambda\mathbf{x}$ for *any* λ . So, we have to put in a qualifier that *the zero vector is never an eigenvector*. Please keep in mind that the number 0 can certainly be an eigenvalue.

Questions

- Find Eigen Values and Particular EigenVector

$$(a) \mathbf{A} = \begin{pmatrix} 7 & 3 \\ 0 & -4 \end{pmatrix}$$

$$(b) \mathbf{A} = \begin{pmatrix} 5 & -2 \\ 4 & -1 \end{pmatrix}$$

Questions

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 1 & 1 \\ 1 & 1 & 2 \end{pmatrix}$$

Some Facts About Eigenvectors

First, some bad news. We cannot use elimination to calculate eigenvalues. Sorry. If we use elimination to convert a matrix A into an upper triangular matrix U , the eigenvalues of U could be different than the eigenvalues of A . However, the two will not be completely unrelated.

What relates them is the amazing fact that the determinant of a matrix is equal to the product of its eigenvalues. That is to say, if $\lambda_1, \dots, \lambda_n$ are the eigenvalues of a matrix A , then

$$\det(A) = \lambda_1 \lambda_2 \cdots \lambda_n$$

So, because A and U have the same determinant, the product of their eigenvalues will be the same.

Some Facts About Eigenvectors

Finally, we note that the trace of a matrix is defined as being the sum of the diagonal elements.

$$\text{trace}(A) = a_{11} + a_{22} + \cdots + a_{nn}$$

The trace of a matrix will be equal to the sum of the eigenvalues of the matrix

$$\text{trace}(A) = \lambda_1 + \lambda_2 + \cdots + \lambda_n.$$

Diagonalizing a Matrix

Diagonalization Suppose the n by n matrix A has n linearly independent eigenvectors $\mathbf{x}_1, \dots, \mathbf{x}_n$. Put those $\mathbf{x}_i$ into the columns of an invertible *eigenvector matrix* X . Then $X^{-1}AX$ is the diagonal *eigenvalue matrix* Λ :

Eigenvector matrix X
Eigenvalue matrix Λ

$$X^{-1}AX = \Lambda = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}. \quad (1)$$

The matrix A is “diagonalized.” We use capital lambda for the eigenvalue matrix, because the small λ ’s (the eigenvalues) are on its diagonal.

Purpose

- The main purpose of diagonalizing a matrix is to simplify complex linear transformations and heavy matrix computations by converting them into a diagonal matrix, where all off-diagonal elements are zero.
- The remarkable relationship between a diagonalized matrix, eigenvalues, and eigenvectors follows from the beautiful mathematical identity (the eigen decomposition) that a square matrix A can be decomposed into the very special form

$$A = P D P^{-1},$$

Diagonalizing a Matrix

Example 1 This A is triangular so its eigenvalues are on the diagonal: $\lambda = 2$ and $\lambda = 6$.

Eigenvectors	$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$	$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$
go into X		

$$\begin{matrix} \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} & \begin{bmatrix} 2 & 4 \\ 0 & 6 \end{bmatrix} & \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} & = & \begin{bmatrix} 2 & 0 \\ 0 & 6 \end{bmatrix} \\ X^{-1} & A & X & = & \Lambda \end{matrix}$$

In other words $A = X\Lambda X^{-1}$. Then watch $A^2 = X\Lambda X^{-1}X\Lambda X^{-1}$. So A^2 is $X\Lambda^2 X^{-1}$.

A^2 has the same eigenvectors in X . It has squared eigenvalues 4 and 36 in Λ^2 .

Diagonalizing a Matrix

we want to express the matrix as a product of three matrices in the form:

$$A = S\Lambda S^{-1}$$

where Λ is a diagonal matrix. In particular, the diagonal entries of Λ will be the eigenvalues of A , and the columns of S will be the corresponding eigenvectors. Having A in this form can greatly simplify many calculations, particularly calculations involving powers of A .

Diagonalizing a Matrix

$$AX = X\Lambda \quad \text{is} \quad X^{-1}AX = \Lambda \quad \text{or} \quad A = X\Lambda X^{-1}. \quad (2)$$

The matrix X has an inverse, because its columns (the eigenvectors of A) were assumed to be linearly independent. *Without n independent eigenvectors, we can't diagonalize A .*

A and Λ have the same eigenvalues $\lambda_1, \dots, \lambda_n$. The eigenvectors are different. The job of the original eigenvectors $\mathbf{x}_1, \dots, \mathbf{x}_n$ was to diagonalize A . Those eigenvectors in X produce $A = X\Lambda X^{-1}$. You will soon see their simplicity and importance and meaning. The k th power will be $A^k = X\Lambda^k X^{-1}$ which is easy to compute using $X^{-1}X = I$:

$$A^k = (X\Lambda X^{-1})(X\Lambda X^{-1}) \dots (X\Lambda X^{-1}) = X\Lambda^k X^{-1}.$$

Things to Remember

Remark 1 Suppose the eigenvalues are n different numbers (like 2 and 6). Then it is automatic that the n eigenvectors will be independent. The eigenvector matrix X will be *invertible*. *Any matrix that has no repeated eigenvalues can be diagonalized.*

Remark 2 *We can multiply eigenvectors by any nonzero constants.* $A(c\mathbf{x}) = \lambda(c\mathbf{x})$ is still true. In Example 1, we can divide $\mathbf{x} = (1, 1)$ by $\sqrt{2}$ to produce a *unit vector*.

MATLAB and virtually all other codes produce eigenvectors of length $\|\mathbf{x}\| = 1$.

Remark 3 The eigenvalues in Λ come in the same order as the eigenvectors in X . To reverse the order of 2 and 6 in Λ , put the eigenvector $(1, 1)$ before $(1, 0)$ in X :

$$X^{-1}AX = \Lambda_{\text{new}} \quad \begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 2 & 4 \\ 0 & 6 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 6 & 0 \\ 0 & 2 \end{bmatrix} = \Lambda_{\text{new}}$$

Things to Remember

Remark 4 (repeated warning for repeated eigenvalues) Some matrices have too few eigenvectors. *Those matrices cannot be diagonalized.* Here are two examples :

Not diagonalizable $A = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}$ **and** $B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$

Their eigenvalues happen to be 0 and 0. Nothing is special about $\lambda = 0$, the problem is the repetition of λ . All eigenvectors of the first matrix are multiples of $(1, 1)$:

**Only one line
of eigenvectors** $A\mathbf{x} = 0\mathbf{x}$ means $\begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ and $\mathbf{x} = c \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$

There is no second eigenvector, so this unusual matrix A cannot be diagonalized.

Those matrices are the best examples to test any statement about eigenvectors. In many true-false questions, non-diagonalizable matrices lead to *false*.

Remember that there is no connection between invertibility and diagonalizability:

1 The columns of $AX = X\Lambda$ are $Ax_k = \lambda_k x_k$. The eigenvalue matrix Λ is diagonal.

2 n independent eigenvectors in X diagonalize A $A = X\Lambda X^{-1}$ and $A^k = X\Lambda^k X^{-1}$

3 Solve $u_{k+1} = Au_k$ by $u_k = A^k u_0 = X\Lambda^k X^{-1}u_0 =$ $c_1(\lambda_1)^k x_1 + \cdots + c_n(\lambda_n)^k x_n$

4 No equal eigenvalues $\Rightarrow$ eigenvector matrix X is invertible and A can be diagonalized.

Repeated eigenvalue $\Rightarrow A$ might have too few independent eigenvectors. Then X^{-1} fails.

5 Every matrix $C = B^{-1}AB$ has the same eigenvalues as A . These C 's are "similar" to A .

Similar Matrices : Same Eigenvalues

Suppose the eigenvalue matrix Λ is fixed. As we change the eigenvector matrix X , we get a whole family of different matrices $A = X\Lambda X^{-1}$ —*all with the same eigenvalues in Λ* . All those matrices A (with the same Λ) are called **similar**.

This idea extends to matrices that can't be diagonalized. Again we choose one constant matrix C (not necessarily Λ). And we look at the whole family of matrices $A = BCB^{-1}$, allowing all invertible matrices B . Again those matrices A and C are called **similar**.

• We are using C instead of Λ because C might not be diagonal. We are using B instead of X because the columns of B might not be eigenvectors. We only require that B is invertible—the key fact about similar matrices stays true: **Similar matrices A and C have the same n eigenvalues.**

Fact : Eigenvectors for n different λ 's are independent. Then we can diagonalize A .

Independent x from different λ Eigenvectors that correspond to distinct (all different) eigenvalues are linearly independent. An n by n matrix that has n different eigenvalues (no repeated λ 's) must be diagonalizable.

Question

$$AX = X\Lambda \quad \text{is} \quad X^{-1}AX = \Lambda \quad \text{or} \quad A = X\Lambda X^{-1}.$$

Example Define the 2×2 matrix

$$A = \begin{bmatrix} -1 & 2 \\ 0 & 3 \end{bmatrix}$$

and

$$P = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ 1 & 0 \end{bmatrix}$$

The inverse of P is

$$P^{-1} = \begin{bmatrix} 0 & 1 \\ -2 & 1 \end{bmatrix}$$